

Generation Interconnection System Impact Study Report

For Queue Project Q18-0962

Rathdrum 138 kV

April 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0962. The Interconnection Customer proposes a 67.6 MW battery energy storage facility interconnecting at the Rathdrum 138 kV substation in the Pennsylvania area. The request was received on March 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 262 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$73,203,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Conoco S - Mill Creek 138 kV	Alexander Street - Springville Substation 138 kV line	129.7	182	5.6
Perryville - Conoco S 138 kV line	Alexander St - Springville 138 kV Line	126.5	199	5.6
Waste Management Northern Oaks Lfgte - Perryville 138 kV	Alexander Street - Springville 138 kV	126.5	213	5.6
Rathdrum - Borger 138 kV Line	Conowingo 345/138kV Transformer Bank	123.8	188	17.8
Alexander St - Springville 138 kV	Waste Management Northern Oaks Lfgte - Springville 138 kV	123.7	268	6.2
Springville - Waste Management Northern Oaks Lfgte 138 kV line	Alexander St - Springville 138 kV L	121.7	229	5.6
Conowingo 345-138 kV Auto	Bixby Hill - Tombstone 500 kV Line	112.8	445	16.3
Conowingo - Borger 138 kV line	Conowingo 345-138 kV XFMR	110.9	194	17.8
Wheelabrator Frackville Energy 500/1 kV Auto	Kline Township Cogen Facility - Talenenergy Martins Crk Llc Harwood 345 kV	110.1	961	5.1
Westwood Generation Llc 345/1 kV Transformer	Foster Wheeler Mt Carmel Cogen 500/345 kV Transformer	101.2	1,040	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Conoco S - Mill Cteek 138 kV	Rebuild line with higher-capacity conductor	\$9,785,000	100% (\$9,785,000)
Conoco S - Perryville 138 kV	Rebuild line with higher-capacity conductor	\$3,918,000	100% (\$3,918,000)
Waste Management Northern Oaks Lfgte - Perryville 138 kV	Rebuild line with higher-capacity conductor	\$8,277,000	100% (\$8,277,000)
Rathdrum - Borger 138 kV line	Rebuild line with higher-capacity conductor	\$27,293,000	100% (\$27,293,000)
Alexander St - Springville 138 kV	Rebuild line with higher-capacity conductor	\$7,498,000	100% (\$7,498,000)
Springville - Waste Management Northern Oaks Lfgte 138 kV line	Rebuild line with higher-capacity conductor	\$3,154,000	100% (\$3,154,000)
Conowingo 345-138 kV Auto	Replace transformer with higher-rated unit	\$17,565,000	61% (\$10,765,000)
Conowingo - Borger 138 kV line	Rebuild line with higher-capacity conductor	\$888,000	100% (\$888,000)
Wheelabrator Frackville Energy 500/1 kV Auto	Replace transformer with higher-rated unit	\$27,537,000	4% (\$1,194,000)
Wstwood Generation Llc 345/1kV TX	Replace transformer with higher-rated unit	\$14,111,000	3% (\$431,000)
	Total allocated to project		\$73,203,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.